

Notes: Wurgler (JFE, 2000)
Financial Markets and the Allocation of Capital

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Manufacturing industry data	3
3.2	Financial development variables	3
3.3	Mechanism variables	4
4	Methodology and empirical specifications	4
4.1	Country-specific allocation elasticity	4
4.2	Cross-country determinants of allocation	4
4.3	Instrumental-variable strategy	5
4.4	Growing versus declining industries	5
4.5	Within-year versus between-year allocation	5
5	Identification and interpretation	5
5.1	What the paper can identify	5
5.2	Why value-added growth is used	6
5.3	Reverse causality and timing	6
5.4	Measurement-error concern	6
5.5	Financial-constraint concern	6
5.6	Interpretation of the main coefficient	6
6	Tables and figures	6
6.1	Table 1: Summary statistics of manufacturing investment and industry growth . . .	6
6.2	Table 2: Country estimates of capital-allocation efficiency	7
6.3	Figure 1: Allocation efficiency versus financial development	8
6.4	Tables 3 and 4: Financial development, asymmetry, and decomposition	9
6.5	Table 5 and Appendix: Mechanisms and financial-development measures	10
6.6	Appendix continued: Financial development across countries	11

7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13
8.2	Economic intuition	13
8.3	Incremental contribution revisited	13
8.4	Implications	13
8.5	Limitations	13
8.6	Takeaway	14

1 Big picture

- **Research question.** Do financial markets improve the real allocation of capital across industries? More specifically, do financially developed countries shift investment toward growing industries and away from declining industries more effectively than financially underdeveloped countries?
- **Main idea.** A core function of an economy is to move capital toward sectors with strong investment opportunities and away from sectors with poor prospects. The paper asks whether larger and better financial markets perform this allocative function better.
- **Empirical object.** For each country, the paper estimates how strongly industry investment growth responds to industry value-added growth. The resulting country-specific elasticity is interpreted as an empirical measure of capital-allocation efficiency.
- **Main finding.** Financially developed countries have much higher investment–value-added elasticities. They expand investment more aggressively in industries whose value added is growing and reduce investment more aggressively in industries whose value added is declining.
- **Mechanisms.** The paper links better capital allocation to three channels: more informative stock prices, lower state ownership, and stronger minority investor rights.
- **Important distinction.** Financial development is not mainly associated with a higher overall level of manufacturing investment. Instead, it is associated with better allocation of a given investment flow across sectors.
- **Economic takeaway.** Financial markets are not merely a sideshow to the real economy. They help decide where capital goes, and this allocation channel is one reason finance can matter for growth and productivity.

2 Incremental contribution of the paper

- **From finance-growth correlation to a concrete channel.** Earlier work documents that financial development is correlated with growth. Wurgler identifies a specific real mechanism: financial development improves the allocation of capital across industries.
- **Direct measurement of allocation.** Instead of inferring allocation from productivity growth or aggregate investment, the paper directly measures whether investment increases in growing industries and decreases in declining industries.
- **Large international industry panel.** The analysis uses a 65-country, 28-industry, 33-year manufacturing panel. This provides broad cross-country variation in both financial development and capital-allocation outcomes.
- **Country-level elasticity as a summary statistic.** The paper turns each country’s industry-year panel into a single interpretable number: the elasticity of investment growth to value-added growth. This becomes a cross-country measure of allocation quality.
- **Beyond total investment.** The paper shows that financially developed countries do not necessarily invest more in manufacturing overall. They allocate investment better.

- **Mechanism evidence.** Wurgler connects allocation quality to stock-price informativeness, state ownership, and minority investor rights, giving empirical support to several theories of how finance affects real activity.
- **Governance insight.** Strong minority investor rights appear especially important for preventing overinvestment in declining industries, consistent with free-cash-flow theories of inefficient investment.

3 Data and measurement

3.1 Manufacturing industry data

The main data source is the United Nations General Industrial Statistics panel, INDSTAT-3. The panel reports gross fixed capital formation, value added, and output for up to 28 three-digit ISIC manufacturing industries across countries and years.

- **Countries.** The final sample contains 65 nonsocialist countries.
- **Years.** The available sample spans 1963–1995, although each country has its own data range.
- **Industries.** The data cover up to 28 manufacturing industries, roughly comparable to two-digit SIC sectors.
- **Investment.** Gross fixed capital formation is used as the industry investment measure.
- **Value added.** Value added is output minus intermediate input costs, with inventory adjustments.
- **Sample cleaning.** Observations are dropped when the absolute value of log investment growth or log value-added growth exceeds one, and very small industries are excluded if they account for less than 0.1 percent of a country’s manufacturing value added in that year.

Interpretation. The dataset is not designed for a structural investment model with capital stocks, but it is broad enough to compare how investment responds to industry growth opportunities across many financial systems.

3.2 Financial development variables

The paper measures financial development using the size of domestic financial markets relative to GDP.

- **Stock-market development:** stock market capitalization to GDP, denoted STK/GDP .
- **Credit-market development:** domestic credit to GDP, denoted $CRED/GDP$.
- **Overall financial development:** FD , the log of one plus the sum of stock market capitalization and credit relative to GDP.
- These variables are based on values around 1980, 1985, and 1990, averaged to smooth cyclical variation.

Interpretation. The paper follows the standard finance-growth literature by using quantities of finance as proxies for the cost and availability of external capital.

3.3 Mechanism variables

The paper then examines more specific channels:

- **Stock-price synchronicity.** From Morck, Yeung, and Yu, this measures the fraction of stocks moving in the same direction in a week. Higher synchronicity means less firm-specific information in stock prices.
- **State ownership.** The share of nonagricultural GDP produced by state-owned enterprises.
- **Investor rights.** An index combining shareholder and creditor rights with the rule of law, based on La Porta et al.

Interpretation. These variables map to three theories: informative prices, privatization/governance, and legal protection of external investors.

4 Methodology and empirical specifications

4.1 Country-specific allocation elasticity

For each country c , Wurgler estimates:

$$\ln \left(\frac{I_{ict}}{I_{ict-1}} \right) = \alpha_c + \eta_c \ln \left(\frac{V_{ict}}{V_{ict-1}} \right) + \varepsilon_{ict}, \quad (1)$$

where I_{ict} is gross fixed capital formation in industry i , country c , and year t , and V_{ict} is value added.

The coefficient η_c is the main measure of capital-allocation efficiency.

- If η_c is high, investment rises strongly in growing industries and falls strongly in declining industries.
- If η_c is near zero, investment growth is weakly related to industry growth opportunities.
- If η_c is negative, investment tends to move in the wrong direction, though almost no countries have significantly negative estimates.

4.2 Cross-country determinants of allocation

The second-stage regression relates η_c to financial development:

$$\eta_c = a + bFD_c + \Gamma'X_c + u_c. \quad (2)$$

Alternative specifications replace FD_c with STK/GDP_c and $CRED/GDP_c$, or add 1960 per capita GDP as a control.

4.3 Instrumental-variable strategy

To address potential reverse causality, the paper instruments financial development with legal origin dummies: English, French, German, and Scandinavian.

The logic is:

- legal origin shapes investor protection;
- investor protection shapes the development of financial markets;
- legal origin is historically predetermined relative to the post-1960 manufacturing allocation outcomes.

Interpretation. The IV estimates are not smaller than OLS; if anything, they are larger. This weakens the view that the finance-allocation relationship is only reverse causality from efficient allocation to larger financial markets.

4.4 Growing versus declining industries

The paper estimates the same country-specific elasticity separately for:

- industry-year observations where value added is growing; and
- industry-year observations where value added is declining.

This answers whether financially developed countries are better at funding winners, cutting losers, or both.

4.5 Within-year versus between-year allocation

The overall elasticity is decomposed into:

- **between-year allocation:** whether aggregate manufacturing investment rises in years when aggregate manufacturing value added rises;
- **within-year allocation:** whether, within a year, investment flows toward industries growing more than other industries.

Interpretation. The decomposition separates market-timing-like allocation from cross-industry portfolio-selection-like allocation.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether investment is more responsive to industry growth opportunities in countries with larger and stronger financial systems. The evidence is strongest as a cross-country correlation supported by several robustness checks and an IV strategy.

5.2 Why value-added growth is used

Value-added growth is used as the measure of industry growth opportunities because comparable international Tobin's Q measures are unavailable. The paper validates this proxy using U.S. data: industry value-added growth is positively correlated with average Q, price-earnings ratios, and sales growth.

5.3 Reverse causality and timing

One concern is that investment may contemporaneously raise value added. Wurgler argues that fixed capital investment typically has gestation lags, so it is unlikely to affect value added immediately in the same year.

5.4 Measurement-error concern

Cross-country data quality differs. Measurement error in value-added growth could attenuate elasticities. However, Wurgler argues that measurement error would have to be extremely large to explain the observed differences across countries.

5.5 Financial-constraint concern

Another concern is that value-added growth might proxy for internal cash flow rather than investment opportunities. The country ranking does not fit this explanation: financially developed countries such as Germany and the United States have high elasticities, while more financially constrained countries often have low elasticities.

5.6 Interpretation of the main coefficient

The elasticity η_c should not be read as a structural parameter from a complete model of investment. It is a transparent reduced-form statistic: a higher value means investment is more strongly reallocated toward expanding industries.

6 Tables and figures

6.1 Table 1: Summary statistics of manufacturing investment and industry growth

Read:

- Table 1 reports country-level data coverage and summary statistics for investment-output ratios, investment growth, and value-added growth.
- The full sample contains 25,201 industry-year observations.
- The average manufacturing investment-output ratio is 6.9 percent.

- Industry investment growth and value-added growth are volatile, especially in developing countries.
- The United States has relatively stable industry investment and value-added growth compared with many other countries.

Economic message: The table documents the broad cross-country and cross-industry variation that makes the empirical strategy possible. It also shows why logs and outlier trimming are important.

Table 1
Summary statistics of total manufacturing investment, industry investment growth, and industry value added growth

Summary statistics for the total manufacturing investment-output ratio, log industry investment growth, and log industry value added growth for 65 non-socialist countries. ISIC-3 industry-year data on gross fixed capital formation, output, and value added are from the 1997 United Nations' *INDSTAT-3* database. Countries report nominal data for up to 28 ISIC-3 manufacturing industries per year. The third column indicates the first and last years for which useful data are available, but for some countries there are interruptions within this range. Values are converted to current U.S. dollars using the year-average exchange rate reported by the IMF's *International Financial Statistics*. Real gross capital formation in industry i in country c in year t (I_{ict}) is then computed by deflating the nominal series by the U.S. capital goods producer price index (base year 1982). Real value added (V_{ict}) and real output are computed by deflating the nominal series by the U.S. finished goods producer price index (base year 1982). The mean manufacturing investment-output ratio is the ratio of total manufacturing investment to total manufacturing output, averaged across years. Observations with absolute values exceeding one for either log investment growth or log value added growth, and those that constitute less than 0.1% of the country's total manufacturing value added in that year, are excluded.

Country	N	Data range	Mean mfg. investment-output ratio	$\ln I_{ict}/I_{ict-1}$		$\ln V_{ict}/V_{ict-1}$	
				Mean	SD	Mean	SD
Australia	526	1963-85	0.043	-0.009	0.327	0.015	0.123
Austria	686	1969-94	0.059	0.041	0.316	0.042	0.162
Bangladesh	180	1981-92	0.033	0.051	0.479	0.066	0.302
Barbados	149	1970-94	0.040	-0.002	0.506	0.030	0.325
Belgium	510	1963-95	0.053	0.029	0.330	0.043	0.152
Bolivia	193	1970-94	0.069	0.109	0.487	0.083	0.273
Cameroon	87	1976-94	0.074	0.046	0.516	0.018	0.380
Canada	670	1963-90	0.044	0.034	0.311	0.031	0.111
Chile	500	1964-94	0.059	0.020	0.481	0.022	0.238
Colombia	598	1963-94	0.037	0.029	0.461	0.050	0.183
Cyprus	472	1971-95	0.056	0.006	0.444	0.067	0.203
Denmark	604	1963-91	0.044	0.044	0.359	0.040	0.152
Ecuador	579	1963-94	0.098	0.066	0.463	0.047	0.269
Egypt	434	1967-93	0.147	-0.038	0.520	0.032	0.356
El Salvador	117	1978-85	0.036	-0.049	0.516	-0.037	0.325
Ethiopia	179	1965-89	0.134	-0.031	0.530	0.053	0.183
Fiji	197	1970-92	0.050	-0.021	0.514	0.039	0.282
Finland	723	1963-94	0.062	0.021	0.401	0.039	0.176
France	325	1963-95	0.064	0.027	0.223	0.052	0.134
Germany	631	1964-92	0.048	0.040	0.216	0.040	0.132
Greece	629	1963-92	0.066	0.012	0.416	0.055	0.166
Guatemala	198	1974-88	0.027	-0.024	0.499	-0.006	0.330
Hong Kong	318	1975-93	0.036	0.045	0.421	0.042	0.181
India	413	1977-93	0.066	0.058	0.369	0.020	0.197
Indonesia	434	1970-95	0.085	0.036	0.454	0.140	0.296

Table 1 (continued)

Country	N	Data range	Mean mfg. investment-output ratio	$\ln I_{ict}/I_{ict-1}$		$\ln V_{ict}/V_{ict-1}$	
				Mean	SD	Mean	SD
Iran	302	1963-93	0.073	-0.012	0.480	0.013	0.367
Ireland	550	1963-91	0.043	0.034	0.404	0.049	0.159
Israel	431	1963-94	0.053	0.065	0.404	0.049	0.170
Italy	522	1967-91	0.055	0.052	0.245	0.045	0.176
Japan	814	1963-93	0.047	0.054	0.276	0.080	0.140
Jordan	263	1974-94	0.362	0.010	0.400	0.080	0.287
Kenya	61	1967-71	0.085	0.054	0.491	0.122	0.152
Korea (South)	682	1966-94	0.092	0.116	0.425	0.148	0.188
Kuwait	290	1968-94	0.080	0.024	0.508	0.062	0.349
Libya	99	1964-80	0.365	0.065	0.450	0.060	0.333
Macao	145	1978-93	0.003	-0.020	0.539	0.093	0.300
Malawi	172	1964-94	0.069	-0.025	0.527	0.012	0.365
Malaysia	334	1968-94	0.064	0.159	0.402	0.150	0.192
Malta	390	1963-93	0.041	-0.027	0.485	0.062	0.226
Mexico	362	1970-91	0.043	0.053	0.428	0.041	0.229
Morocco	108	1985-94	0.060	0.094	0.405	0.078	0.163
Netherlands	616	1963-93	0.047	0.031	0.292	0.043	0.139
New Zealand	377	1963-90	0.043	0.013	0.388	0.038	0.153
Nigeria	161	1963-90	0.102	0.018	0.493	0.093	0.358
Norway	717	1963-92	0.053	0.035	0.379	0.025	0.156
Pakistan	176	1965-91	0.087	-0.032	0.438	0.078	0.283
Panama	333	1963-91	0.054	-0.018	0.504	0.041	0.215
Peru	201	1982-92	0.031	0.009	0.492	0.032	0.366
Philippines	527	1963-93	0.047	0.035	0.469	0.041	0.306
Portugal	557	1971-94	0.066	0.034	0.409	0.047	0.197
Singapore	642	1963-94	0.053	0.074	0.424	0.099	0.194
Spain	563	1964-92	0.039	0.041	0.329	0.046	0.159
Sri Lanka	126	1979-93	0.049	0.050	0.551	-0.021	0.341
Swaziland	53	1970-90	0.127	-0.013	0.517	0.012	0.329
Sweden	565	1963-87	0.062	0.010	0.297	0.031	0.139
Tanzania	220	1965-91	0.086	-0.049	0.479	0.042	0.314
Trinidad & Tobago	73	1967-91	0.044	-0.023	0.531	-0.004	0.282
Tunisia	387	1963-95	0.074	0.047	0.437	0.087	0.215
Turkey	596	1963-94	0.052	0.038	0.434	0.062	0.259
United Kingdom	620	1968-91	0.040	0.018	0.263	0.027	0.142
United States	868	1963-95	0.034	0.027	0.186	0.022	0.091
Uruguay	85	1989-93	0.027	-0.009	0.468	0.038	0.227
Venezuela	352	1976-93	0.065	-0.051	0.457	-0.029	0.273
Zambia	106	1963-75	0.070	-0.009	0.498	0.132	0.257
Zimbabwe	403	1963-94	0.052	0.019	0.456	0.027	0.197
Full Sample	25,201	1963-95	0.069	0.030	0.402	0.050	0.217
				(equal-wtd.)	(N-wtd.)	(N-wtd.)	(N-wtd.)

Table 1, part 1: Summary statistics by country.

Table 1, part 2: Continued summary statistics.

6.2 Table 2: Country estimates of capital-allocation efficiency

Read:

- Table 2 reports the country-specific elasticity η_c from the investment-growth/value-added-growth regression.
- Germany has the highest estimated elasticity, 0.988, followed by Hong Kong and New Zealand.
- Japan, the United Kingdom, and the United States all rank in the top quintile of the 65-country sample.

- Some countries have elasticities close to zero, such as Panama, Kuwait, Malawi, and India.
- The mean country elasticity is 0.429, with a cross-country standard deviation of 0.288.

Economic message: Countries differ greatly in how strongly investment follows industry growth opportunities. The high rankings of Germany, Japan, the UK, and the US suggest that very different developed financial systems can all support efficient allocation.

Table 2

Estimates of the elasticity of industry investment to value added

Estimates of the elasticity of industry investment to industry value added in 65 nonsociali countries. The estimates are obtained from the following regression, estimated for each country i

$$\ln \frac{I_{ict}}{I_{ict-1}} = \alpha_c + \eta_c \ln \frac{V_{ict}}{V_{ict-1}} + v_{ict},$$

where i indexes ISIC-3 manufacturing industries, c indexes countries, and t indexes years. Th number of observations in each country is reported in Table 1. Robust standard errors are i parentheses.

Country	$\hat{\eta}_c$ (se)	R^2	Rank of $\hat{\eta}_c$
Australia	0.681 (0.134)	0.065	14
Austria	0.835 (0.085)	0.182	8
Bangladesh	0.131 (0.120)	0.007	54
Barbados	0.072 (0.123)	0.002	60
Belgium	0.803 (0.098)	0.137	11
Bolivia	− 0.202 (0.128)	0.013	65
Cameroon	0.134 (0.149)	0.010	53
Canada	0.547 (0.115)	0.038	26
Chile	0.294 (0.100)	0.021	40
Colombia	0.130 (0.108)	0.006	55
Cyprus	0.421 (0.117)	0.037	30
Denmark	0.853 (0.123)	0.131	6
Ecuador	0.305 (0.071)	0.031	39
Egypt	0.326 (0.069)	0.050	36
El Salvador	0.262 (0.165)	0.027	46
Ethiopia	0.135 (0.247)	0.002	52
Fiji	0.154 (0.133)	0.007	51
Finland	0.557 (0.087)	0.059	25
France	0.893 (0.075)	0.289	4
Germany	0.988 (0.061)	0.364	1
Greece	0.635 (0.104)	0.064	20
Guatemala	0.633 (0.091)	0.176	21
Hong Kong	0.948 (0.132)	0.166	2
India	0.100 (0.097)	0.003	57
Indonesia	0.217 (0.077)	0.020	50
Iran	0.446 (0.067)	0.116	29
Ireland	0.666 (0.114)	0.069	15
Israel	0.263 (0.107)	0.012	45
Italy	0.652 (0.063)	0.220	16
Japan	0.819 (0.074)	0.174	9
Jordan	0.322 (0.096)	0.053	37
Kenya	0.068 (0.389)	0.000	61
Korea (South)	0.646 (0.089)	0.082	18

Table 2 (continued)

Country	$\hat{\eta}_c$ (se)	R^2	Rank of $\hat{\eta}_c$
Kuwait	0.047 (0.087)	0.001	63
Libya	0.387 (0.122)	0.082	31
Macao	0.237 (0.147)	0.017	32
Malawi	0.075 (0.115)	0.003	59
Malaysia	0.285 (0.118)	0.019	42
Malta	0.268 (0.102)	0.016	44
Mexico	0.344 (0.114)	0.034	34
Morocco	0.638 (0.227)	0.066	19
Netherlands	0.573 (0.093)	0.074	24
New Zealand	0.896 (0.130)	0.125	3
Nigeria	0.364 (0.106)	0.070	33
Norway	0.575 (0.093)	0.056	23
Pakistan	0.255 (0.130)	0.027	47
Panama	0.064 (0.125)	0.001	62
Peru	0.651 (0.081)	0.234	17
Philippines	0.313 (0.075)	0.042	38
Portugal	0.539 (0.097)	0.068	27
Singapore	0.486 (0.088)	0.049	28
Spain	0.867 (0.077)	0.175	5
Sri Lanka	0.273 (0.156)	0.029	43
Swaziland	− 0.069 (0.217)	0.002	64
Sweden	0.852 (0.083)	0.159	7
Tanzania	0.087 (0.102)	0.003	58
Trinidad & Tobago	0.340 (0.250)	0.032	35
Tunisia	0.287 (0.116)	0.020	41
Turkey	0.242 (0.072)	0.021	48
United Kingdom	0.812 (0.092)	0.192	10
United States	0.723 (0.069)	0.126	13
Uruguay	0.218 (0.257)	0.011	49
Venezuela	0.593 (0.082)	0.125	22
Zambia	0.123 (0.182)	0.004	56
Zimbabwe	0.726 (0.116)	0.099	12
Mean	0.429	0.071	(Out of 65)
Median	0.344	0.042	
SD	0.288	0.078	

Table 2, part 1: Estimated elasticities by coun- try.

Table 2, part 2: Continued estimated elasticities.

6.3 Figure 1: Allocation efficiency versus financial development

Read:

- Figure 1 plots the country-specific allocation elasticity against the financial development index FD .
- The fitted line slopes upward, showing a strong positive relationship between financial development and capital-allocation efficiency.
- Financially developed countries such as Japan, Germany, the UK, and the US are toward the upper-right portion of the figure.

- Countries with weaker financial development, such as Bolivia, Swaziland, Panama, and Kuwait, appear near the lower-left region.

Economic message: This is the central visual result. More financially developed countries allocate capital more responsively across industries.

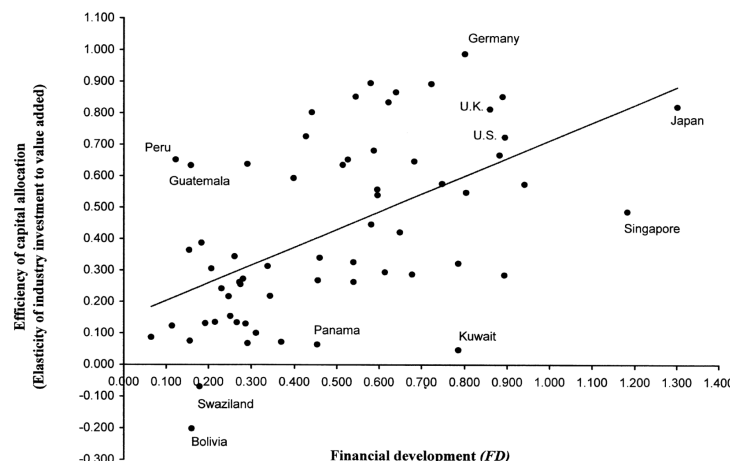


Figure 1: Efficiency of capital allocation versus financial development.

6.4 Tables 3 and 4: Financial development, asymmetry, and decomposition

Read Table 3:

- The dependent variable is the country-specific allocation elasticity from Table 2.
- The financial development index FD is strongly positively associated with allocation efficiency.
- Stock-market capitalization and credit-market size are individually positively related to allocation efficiency.
- When stock and credit variables are included together, credit-market size dominates.
- The relationship remains after controlling for 1960 per capita GDP.
- Legal-origin IV estimates are also positive, supporting the view that financial development has an independent effect.

Read Table 4:

- Table 4 separates allocation in growing industries from allocation in declining industries.
- Financial development is associated with higher investment sensitivity in both growing and declining industries.
- The result suggests that financially developed countries both fund winners and cut losers.
- The table also separates between-year and within-year components.

- Financial development is positively related to both components, but stock markets appear more connected to within-year allocation while credit markets are more connected to between-year allocation.

Economic message: Financial development improves allocation on multiple margins: across time, across industries within a year, toward growing sectors, and away from declining sectors.

Table 3

The allocation of capital and broad measures of financial development

The dependent variable is the estimated elasticity of manufacturing investment to value added from Table 2. The independent variables are a summary measure of financial development (FD_t), a measure of stock market capitalization to GDP (STK/GDP_t), a measure of credit outstanding to GDP ($CRED/GDP_t$), and the 1960 value of log per capita GDP. These variables are summarized in the Appendix. In specifications (1)-(7), estimation is by ordinary least-squares. In specifications (8) and (9), legal origin dummies (English, French, German, or Scandinavian) from La Porta et al. (1998) are used as instruments for the financial development measures (first stage R^2 between 0.23 and 0.30, F -statistics significant at 1%). Robust standard errors are in parentheses. ***, **, * and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent variable: β_t	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8) (I.V.)	(9) (I.V.)
FD_t	0.565*** (0.107)					0.323*** (0.105)		0.637** (0.276)	
STK/GDP_t		0.718*** (0.175)		0.227 (0.247)			0.186 (0.160)		0.038 (0.370)
$CRED/GDP_t$			0.736*** (0.132)	0.614*** (0.187)			0.304* (0.157)		0.872*** (0.322)
GDP_{1960}					0.211*** (0.024)	0.154*** (0.030)	0.157*** (0.030)	0.114* (0.059)	0.109* (0.059)
Intercept	0.147** (0.058)	0.319*** (0.038)	0.131** (0.063)	0.146** (0.066)	0.289*** (0.033)	0.163*** (0.054)	0.171*** (0.060)	0.059 (0.118)	0.030 (0.115)
R^2	0.307	0.226	0.289	0.301	0.432	0.519	0.517	0.380	0.374
N	63	65	63	63	62	61	61	45	45

Table 3: Allocation of capital and broad measures of financial development.

Table 4

Capital allocation in growing vs. declining industries and between-year vs. within-year

The dependent variable in (1)-(3) is the elasticity of manufacturing investment to value added, estimated using only observations in which industry value added increased versus the prior year. The dependent variable in (4)-(6) uses only observations in which industry value added decreased versus the prior year. The dependent variable in (7)-(9) is the between-year estimator (for a given country, a regression of mean industry investment growth in that year on mean industry value added growth in that year). The sample in (7)-(9) includes only countries with at least 15 years of data. The dependent variable in (10)-(12) is the within-year estimator (the year fixed-effects estimator). The independent variables are a summary measure of financial development (FD_t), a measure of stock market capitalization to GDP (STK/GDP_t), a measure of credit outstanding to GDP ($CRED/GDP_t$), and the 1960 value of log per capita GDP. These variables are summarized in the Appendix. Estimation is by ordinary least-squares. Robust standard errors are in parentheses. ***, **, * and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent variable:	$\hat{\beta}_t^+$			$\hat{\beta}_t^-$			$\hat{\beta}_t^b$			$\hat{\beta}_t^w$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
FD_t	0.571*** (0.130)			0.268* (0.162)	0.577*** (0.172)		0.449** (0.179)	0.551** (0.215)		0.425** (0.196)	0.278*** (0.081)	0.213** (0.082)
STK/GDP_t		0.077 (0.264)			0.071 (0.308)			-0.010 (0.380)			0.293* (0.150)	
$CRED/GDP_t$			0.746*** (0.245)		0.745*** (0.279)			0.773** (0.331)			0.166 (0.129)	
GDP_{1960}				0.178*** (0.046)		0.093 (0.071)			0.088 (0.102)		0.053* (0.030)	
Intercept	0.058 (0.077)	0.030 (0.090)	0.088 (0.078)	0.088 (0.101)	0.043 (0.112)	0.069 (0.103)	0.419*** (0.146)	0.382** (0.152)	0.430*** (0.145)	0.095* (0.052)	0.122** (0.057)	0.091* (0.051)
R^2	0.219	0.232	0.386	0.155	0.160	0.206	0.109	0.118	0.147	0.138	0.139	0.208
N	63	63	61	63	63	61	55	55	53	63	63	61

Table 4: Growing vs. declining industries and within-year vs. between-year allocation.

6.5 Table 5 and Appendix: Mechanisms and financial-development measures

Read Table 5:

- Stock-price synchronicity is negatively related to allocation efficiency.
- Since lower synchronicity means more firm-specific information in stock prices, this supports the view that informative stock prices guide investment.
- State ownership is negatively related to allocation efficiency.
- Investor rights are positively related to allocation efficiency.
- Investor rights are especially associated with the ability to keep investment out of declining industries.

Read Appendix:

- The appendix reports the country-level financial-development variables used in the cross-country regressions.
- Japan, Singapore, Malaysia, Sweden, the UK, and the US appear among the high financial-development countries.
- Low financial-development countries include Tanzania, Zambia, Nigeria, Peru, Guatemala, and Malawi.

Economic message: The mechanism evidence links capital allocation not only to financial market size but also to information, ownership, and investor protection.

Table 5
Capital allocation and the synchronicity of stock prices, state ownership, and investor rights

The dependent variable in (1)–(6) is the estimated elasticity of manufacturing investment to value added from Table 2. The dependent variable in (7) and (8) is the difference between the elasticity estimate for growing industry-year observations and the estimate for declining industry-year observations. The independent variables are a measure of stock price synchronicity (the average fraction of stocks moving in the same direction in a given week of the first 26 weeks of 1995) from Morck et al. (2000) (mean = 0.664, SD = 0.043, $n = 53$). SOE is the fraction of an economy's output due to state-owned enterprises (mean = 0.132, SD = 0.106, $n = 39$) from the World Bank (1995). $RIGHTS$ is an index of effective investor rights. It is the product of a measure of the rule of law (0–1, continuous) and the number of important shareholder and creditor rights (0–10, integer) in the country's legal code (mean = 3.73, SD = 1.93, $n = 41$). Both variables are from La Porta et al. (1998). Other independent variables are a summary measure of financial development (FD) and the 1960 value of log per capita GDP, as summarized in the Appendix. Estimation is by ordinary least-squares. Robust standard errors are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent variable:	$\hat{\theta}_i$						$\hat{\theta}_i^* - \hat{\theta}_i^{\dagger}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$SYNCH_i$	-3.185*** (0.745)			-3.042*** (0.927)	-2.492*** (0.609)	-1.682** (0.741)	-1.370 (1.895)	
SOE_i		-0.904** (0.369)		-1.904*** (0.596)	-1.824*** (0.500)	-1.094 (0.726)	1.306 (0.982)	
$RIGHTS_i$			0.062*** (0.015)	0.016 (0.023)	-0.033 (0.026)	-0.033 (0.024)	0.100*** (0.028)	0.117*** (0.037)
FD_i					0.580** (0.204)	0.434 (0.276)	-0.503** (0.211)	-0.771 (0.499)
GDP_i						0.121 (0.081)	0.039 (0.137)	
<i>Intercept</i>	2.714*** (0.500)	0.536*** (0.074)	0.315*** (0.070)	2.727*** (0.657)	2.208*** (0.420)	1.557*** (0.526)	0.724 (1.352)	
R^2	0.289	0.103	0.200	0.591	0.680	0.723	0.167	0.358
N	33	39	41	22	22	22	40	22

Table 5: Stock-price synchronicity, state ownership, investor rights, and allocation.

Appendix A. Summary statistics of financial development

Summary statistics of measures of financial development for 65 non-socialist countries. Financial development is computed as the sum of stock market capitalization to GDP and private and nonfinancial public domestic credit to GDP. (The components are averaged over 1980, 1985, and 1990 values, or as available.) The capitalization of stock markets is from the International Finance Corporation's *Emerging Stock Markets Factbook*, and domestic credit is from the *International Financial Statistics Yearbook*. The size of the credit market is the sum of lines 32c and 32d (claims on the private sector and claims on the nonfinancial public sector; these are items in the "Domestic Monetary Survey" category) and lines 52c and 52d as available (items in the "Financial Survey" category), or 42c and 42d (items in the "Other Financial Institutions" category) where 52c and 52d are not available. GDP is the 1960 value of per capita GDP, in thousands of 1960 dollars, from the Penn World Tables version 5.6 (Summers and Heston, 1991).

Country	Financial development, 1980–1990 [FD_e = $\ln(1 + \cdot)$]	Stock market cap. to GDP, 1980–1990 [STK/GDP_e = $\ln(1 + \cdot)$]	Credit claims to GDP, 1980–1990 [$CRED/GDP_e$ = $\ln(1 + \cdot)$]	Per capita GDP, 1960 (\$000) [GDP_e = $\ln(\cdot)$]
Australia	0.80	0.36	0.44	7.75
Austria	0.86	0.09	0.77	5.14
Bangladesh	0.21	0.01	0.20	0.94
Barbados	0.45	0.06	0.39	2.64
Belgium	0.55	0.23	0.32	5.47
Bolivia	0.17	0.00	0.17	1.13
Cameroon	0.31	0.00	0.31	0.63
Canada	1.23	0.43	0.80	7.24
Chile	0.85	0.31	0.54	2.90
Colombia	0.33	0.03	0.30	1.69

Appendix A, part 1: Financial-development summary statistics.

6.6 Appendix continued: Financial development across countries

Read:

- The continuation of the appendix reports country-by-country values of FD , stock-market capitalization to GDP, credit claims to GDP, and 1960 per capita GDP.
- These variables provide the basis for the cross-country explanatory regressions in Tables 3–5.
- The decomposition of FD shows that countries can have high financial development through stock markets, credit markets, or both.

Economic message: The financial-development index captures broad financial depth, but its components also allow Wurgler to separate the roles of equity and credit markets.

Appendix (continued)

Country	Financial development, 1980–1990 [FD_c] = $\ln(1 + \cdot)$]	Stock market cap. to GDP, 1980–1990 [STK/GDP_c] = $\ln(1 + \cdot)$]	Credit claims to GDP, 1980–1990 [$CRED/GDP_c$] = $\ln(1 + \cdot)$]	Per capita GDP, 1960 (\$000) [GDP_c] = $\ln(\cdot)$]
Cyprus	0.91	0.14	0.77	2.08
Denmark	0.72	0.21	0.51	6.73
Ecuador	0.24	0.03	0.21	1.46
Egypt	0.72	0.04	0.68	0.80
El Salvador	0.31	0.00	0.31	1.43
Ethiopia	0.24	0.00	0.24	0.26
Fiji	0.29	0.00	0.29	2.11
Finland	0.81	0.11	0.70	5.28
France	1.06	0.17	0.89	5.82
Germany	1.22	0.21	1.01	6.57
Greece	0.67	0.11	0.56	2.09
Guatemala	0.17	0.00	0.17	1.66
Hong Kong	-	1.21	-	2.23
India	0.36	0.08	0.28	0.77
Indonesia	0.28	0.03	0.25	0.64
Iran	0.79	0.04	0.75	2.99
Ireland	1.42	0.72	0.70	3.30
Israel	0.71	0.23	0.48	3.45
Italy	0.69	0.11	0.58	4.58
Japan	2.67	0.67	2.00	2.94
Jordan	1.19	0.46	0.73	1.16
Kenya	0.34	0.02	0.32	0.65
Korea (South)	0.98	0.20	0.78	0.90
Kuwait	1.20	0.49	0.71	-
Libya	0.20	0.00	0.20	-
Macao	-	0.00	-	-
Malawi	0.17	0.00	0.17	0.38
Malaysia	1.44	0.72	0.72	1.41
Malta	0.58	0.00	0.58	1.38
Mexico	0.29	0.07	0.22	2.83
Morocco	0.34	0.03	0.31	0.83
Netherlands	1.56	0.39	1.17	6.09
New Zealand	0.79	0.30	0.49	7.95
Nigeria	0.16	0.03	0.13	0.56
Norway	1.11	0.15	0.96	5.59
Pakistan	0.32	0.05	0.27	0.64
Panama	0.57	0.01	0.56	1.57
Peru	0.13	0.04	0.09	2.03

Appendix A, part 2: Financial-development data by country.

Appendix (continued)

Country	Financial development, 1980–1990 [FD_c] = $\ln(1 + \cdot)$]	Stock market cap. to GDP, 1980–1990 [STK/GDP_c] = $\ln(1 + \cdot)$]	Credit claims to GDP, 1980–1990 [$CRED/GDP_c$] = $\ln(1 + \cdot)$]	Per capita GDP, 1960 (\$000) [GDP_c] = $\ln(\cdot)$]
Philippines	0.40	0.09	0.31	1.13
Portugal	0.82	0.06	0.76	1.86
Singapore	2.26	1.23	1.03	1.63
Spain	0.90	0.14	0.76	3.13
Sri Lanka	0.31	0.09	0.24	1.25
Swaziland	0.20	0.00	0.20	1.24
Sweden	1.43	0.29	1.14	7.57
Tanzania	0.07	0.00	0.07	0.32
Trinidad & Tobago	0.58	0.12	0.46	5.62
Tunisia	0.97	0.05	0.92	1.10
Turkey	0.26	0.07	0.19	1.62
United Kingdom	1.36	0.66	0.70	6.81
United States	1.44	0.56	0.88	9.91
Uruguay	0.41	0.00	0.41	3.96
Venezuela	0.49	0.08	0.41	6.31
Zambia	0.12	0.00	0.12	0.95
Zimbabwe	0.53	0.23	0.30	1.00
Mean	0.71	0.19	0.52	2.94
SD	0.52	0.27	0.34	2.44
Mean [$\ln(1 + \cdot)$]	0.49	0.15	0.40	Not used
SD [$\ln(1 + \cdot)$]	0.28	0.19	0.21	Not used
Mean [$\ln(\cdot)$]	Not used	Not used	Not used	0.71
SD [$\ln(\cdot)$]	Not used	Not used	Not used	0.90

Appendix A, part 3: Continued country data and summary statistics.

7 Short summary

- The paper studies whether financial markets improve the allocation of capital across industries.
- It uses a 65-country manufacturing panel covering up to 28 industries from 1963 to 1995.
- For each country, it estimates the elasticity of industry investment growth to industry value-added growth.
- A higher elasticity means investment responds more strongly to growth opportunities.
- Financially developed countries have substantially higher elasticities.
- The result holds for both stock-market and credit-market development, though credit-market size is especially strong in some specifications.
- Financial development helps both by increasing investment in growing industries and decreasing investment in declining industries.
- Stock markets matter partly through the information content of prices: less synchronous stock prices are associated with better allocation.

- State ownership is negatively related to allocation efficiency.
- Minority investor rights are positively related to allocation efficiency, especially by limiting overinvestment in declining industries.
- The paper identifies better capital allocation as a concrete channel through which financial markets affect real economic performance.

8 Conclusion

8.1 Core conclusion

Wurgler shows that financial markets improve the allocation of capital. Countries with more developed financial sectors shift investment more strongly toward growing manufacturing industries and away from declining industries. This effect is visible across a large panel of countries, industries, and years.

8.2 Economic intuition

Financial markets help capital flow to where it is most valuable. Informative stock prices help managers and investors identify good projects. Credit markets help finance expansion in industries with strong prospects. Investor rights and governance institutions reduce the ability of managers or governments to keep reinvesting in bad projects. Together, these mechanisms make investment more responsive to real opportunities.

8.3 Incremental contribution revisited

The paper's key contribution is to make the finance-growth relationship concrete. Rather than saying that finance is correlated with growth, Wurgler shows that finance is associated with a better sectoral allocation of investment. This helps explain why financial development matters for productivity and long-run growth.

8.4 Implications

The findings imply that policy discussions about financial development should focus not only on the amount of investment, but also on where investment goes. A country may not raise aggregate investment by developing financial markets, but it may improve economic performance by allocating a given amount of investment more effectively.

8.5 Limitations

The empirical design is cross-country and observational. Financial development may be correlated with other institutions that also improve allocation. The IV strategy using legal origin helps, but it does not eliminate all concerns. The measure of investment opportunities, value-added growth, is imperfect. International manufacturing data also differ in quality across countries.

8.6 Takeaway

The enduring takeaway is that finance matters because it allocates capital. Developed financial markets and investor-protection institutions help economies fund growing industries, discipline declining industries, and reduce politically or managerially distorted investment.